

Deepak Mallampati

AI / MACHINE LEARNING ENGINEER

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US-based · F-1 OPT · Open to remote (EU collaboration)

PROFESSIONAL SUMMARY

AI / Machine Learning Engineer with 4+ years building production ML and LLM systems with MLOps discipline. Trains ML models for prediction and forecasting, integrates LLM-powered features (GPT APIs) into products, and deploys with reproducibility and versioning on AWS - reducing failed production runs by 60% and serving 100,000+ requests daily. Builds and automates AI workflows end-to-end in fast-paced, ownership-driven environments.

CORE COMPETENCIES & SKILLS

Machine Learning (Forecasting & Optimization) • LLM Product Features (GPT APIs) • MLOps & Model Deployment • AWS Cloud Infrastructure • Automation Pipelines • Reproducibility & Versioning • Startup 0 → 1 Delivery

Machine Learning: Regression, Classification, Forecasting | PyTorch, TensorFlow, scikit-learn | Feature Engineering | Model Optimization

Generative AI & LLMs: GPT-4o/5, Claude, Gemini | OpenAI/GPT APIs | RAG | Prompt Engineering | Multi-Agent Systems

MLOps & Cloud: AWS (EC2, S3, Lambda, SageMaker, Bedrock) | Docker, Kubernetes, Terraform | MLflow, W&B | CI/CD (Jenkins, Azure DevOps)

Data Engineering: PostgreSQL, MongoDB, Redis | Spark, Databricks, Kafka | Pandas, NumPy | Data Lineage

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to materially reduce hallucination.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) enabling compliant analytics with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry, reducing failed production runs by **60%**.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** on internal eval datasets.
- Engineered safeguards and content controls, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Delivered scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical readmission dataset (k-anonymity, l-diversity, Laplace differential privacy across four ϵ levels).
- Reduced re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (RF AUC 0.689 vs 0.696).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated 361-example instruction dataset for controllable long-form generation.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Developed supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Engineered features and end-to-end preprocessing pipelines on raw sensor and operational datasets.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens [Chrome Web Store](#)

Shipped Chrome extension delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

career-ops

Autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0